

SONAL SINGH

Software Engineer | Agentic AI & LLM Systems

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SUMMARY

Software Engineer with 2+ years designing and shipping production agentic AI systems. Built **Kalp**, an MCP server publishing 15 tools across six specialized agents, with multi-provider LLM routing (Claude, GPT, Ollama, DeepSeek, NVIDIA NIM, OpenRouter) and a pgvector-backed RAG pipeline serving external clients. Hands-on with LangGraph, LangChain, FastAPI, and MCP for agent orchestration, plus REST/webhook integrations — auth, data mapping, error handling — connecting to third-party CRMs and messaging platforms. Comfortable owning agent reliability end-to-end: prompt design, tool-calling, guardrails, retries, and translating technical work into plans non-technical stakeholders can act on.

TECHNICAL SKILLS

AI / Agentic Systems: LangGraph, LangChain, MCP (Model Context Protocol), FastAPI, multi-agent orchestration, multi-provider LLM routing (Claude/GPT/Gemini/Ollama/DeepSeek/NVIDIA NIM/OpenRouter), prompt engineering, tool/function calling

RAG & Vector Search: RAG pipeline design, pgvector, Qdrant, document-fingerprinting, retrieval optimization

Integrations & Automation: REST/webhook APIs (auth, data mapping, error handling), n8n, JSON-based integrations, third-party integrations (Zoho CRM, WhatsApp Cloud API)

Languages: Python, Java (Core, OOP, Java 21), JavaScript/TypeScript, SQL

Backend & Data: Spring Boot, PostgreSQL, MySQL, Redis, Docker, AWS (EC2, S3), GitHub Actions CI/CD, Jenkins

Reliability & Security: Retries, timeouts, idempotent processing, rollback handling, guardrails/validation for AI outputs, JWT/OAuth2, RBAC

Tools: Git, GitHub, Postman, Claude Code, Cursor, GitHub Copilot

EXPERIENCE

Software Engineer — PetroIT Software Solutions

Feb 2024 – Present

Gurugram, India

- Designed and built **Kalp**, a production MCP server embedded in Mozaic, publishing 15 tools across six specialized agents (reporting, analysis, form-fill, document-linking, LOV management, extraction), with tool-calling, persisted chat memory, and streaming responses.
- Implemented multi-provider LLM routing (Claude, GPT, Ollama, DeepSeek, NVIDIA NIM, OpenRouter) via a custom ModelRouter, plus a pgvector-backed RAG and document-fingerprinting pipeline, deployed via Docker/Jenkins with JWT/OAuth2-secured endpoints for external clients.
- Built and maintained production AI-assisted workflows using Claude, OpenAI, and Gemini APIs, including guardrails and validation logic to keep AI-generated outputs stable in a live system.
- Built a secure RBAC framework with Spring Security and optimized 20+ PostgreSQL queries with indexing strategies, cutting response times 40%.
- Authored and presented an AI adoption roadmap and governance plan to engineering leadership, translating technical capability into a business-readable strategy and 30/60/90-day rollout plan.

PROJECTS

Knowledge-Based Bots — Multi-Agent Engineering Assistant Architecture

2026 – Present

- Designed a multi-agent knowledge-bot architecture, each agent scoped to its own domain/tools/data, unified through a shared orchestration layer, with least-privilege tool access enforced at the middleware layer and a shared vector memory system for secure cross-agent knowledge sharing.

Anuvaad — AI-Powered Multilingual Enterprise Platform

2026 – Present

- Built resilient multi-LLM job orchestration (Mistral, LLaMA, Gemma, GPT-class) with automatic retries, rollback, and idempotent re-processing, plus secure external REST APIs, webhook integrations, RBAC, and real-time SSE progress tracking.

Flight61 — Aviation Safety & Pilot-Training Analytics Platform

Chimes Aviation Academy

- Integrated the Anthropic Claude API to generate plain-English debrief narratives from structured safety data, plus an admin AI chatbot answering natural-language questions against a read-only DB view, with WhatsApp Cloud API and Zoho CRM webhook integrations for automated alerts and data sync.

EDUCATION

B.E.(Computer Science)

Aug 2020 – Jul 2024

Chandigarh University

8.15 CGPA